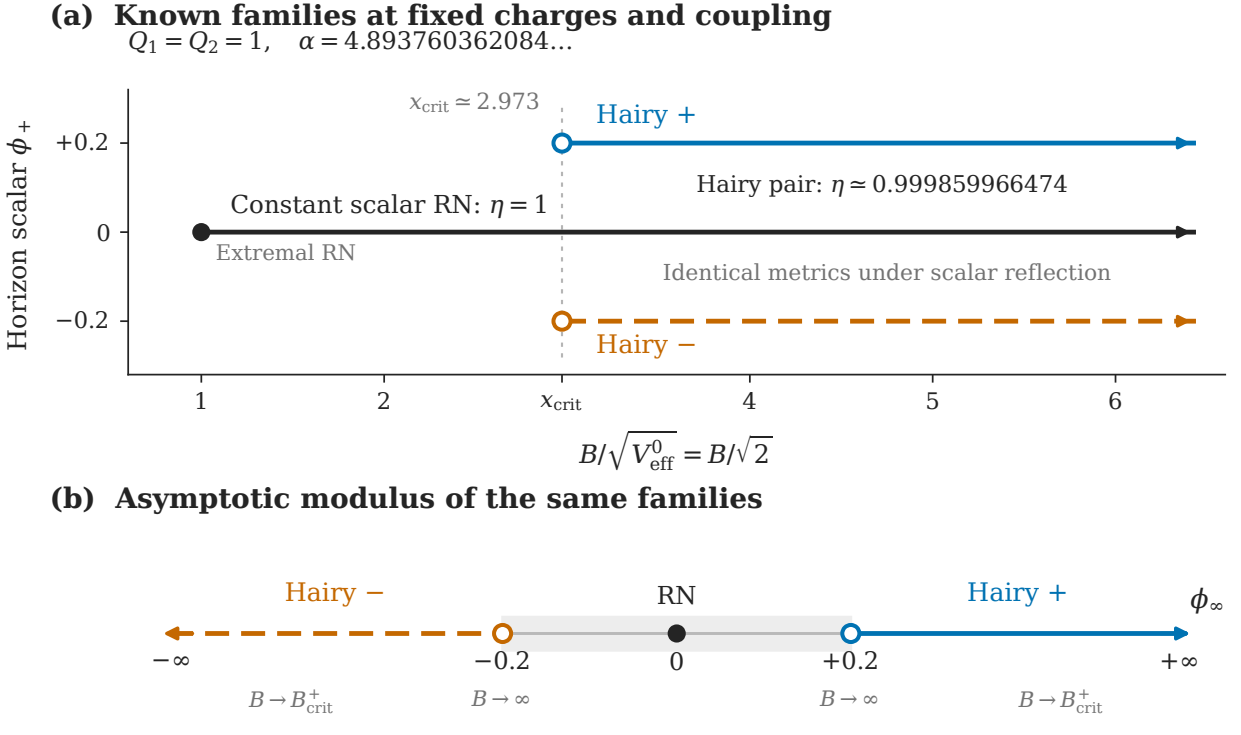
(a) Known families at fixed charges and coupling
$Q_1 = Q_2 = 1, \quad \alpha = 4.893760362084\ldots$
$x_{\rm crit} \simeq 2.973$
Hairy +
Hairy pair: $\eta \simeq 0.999859966474$
Constant scalar RN: $\eta = 1$
Extremal RN
Identical metrics under scalar reflection
Hairy −
Horizon scalar ϕ_+
+0.2
0
−0.2
1
2
$x_{\rm crit}$
4
5
6
$B/\sqrt{V^0_{\rm eff}} = B/\sqrt{2}$
(b) Asymptotic modulus of the same families
Hairy −
RN
Hairy +
ϕ_∞
$-\infty$
−0.2
0
+0.2
$+\infty$
$B \to B^+_{\rm crit}$
$B \to \infty$
$B \to \infty$
$B \to B^+_{\rm crit}$
Grey intervals: not covered by the families shown.